

Teaching Quantum Mechanics and Computing with AI

From abstract concepts to interactive, personalised learning

Taha Selim

t.i.m.m.selim2@hva.nl

**Research/Lecturer of Quantum Computing and Quantum Theoretical Chemistry
Amsterdam University of Applied Sciences**

QuBIT EDU, Sept. 30th, 2026, Aachen
Track B – Hörsaal (H08)

Quantum AI Conference 2026 in Indianapolis

- Introduce the latest advancements in Quantum AI
- Quantum Advantage
- Wide audience from different countries and academic backgrounds;
- Non-quantum experts and researchers from the medical field, data science, informatics, and engineering.



Quantum AI Conference 2026 in Indianapolis

- Introduce the latest advancements in Quantum AI
- Quantum Advantage
- Wide audience from different countries and academic backgrounds;
- Non-quantum experts and researchers from the medical field, data science, informatics, and engineering.



The people behind quantum education HvA



Marten Teitsma

Lector Applied
Quantum
Computing (AQC)



Taha Selim

Educational Officer
QTLC-A
Lecturer/Researcher
HvA



**Samuel van
Beek**

Educational Officer
QTLC-A
Innovation Officer
QuSoft



Ilse Geurts

Project Manager
QTLC-A &
Lectoraat AQC

Curriculum Committee of Applied Quantum Technology Master

**Dr. Marijn
A.M. Versteegh**

Senior Lecturer and
Researcher, The
Hague University
of Applied Sciences

Lotte Spanjers

Lecturer and Researcher,
Fontys University
of Applied Sciences



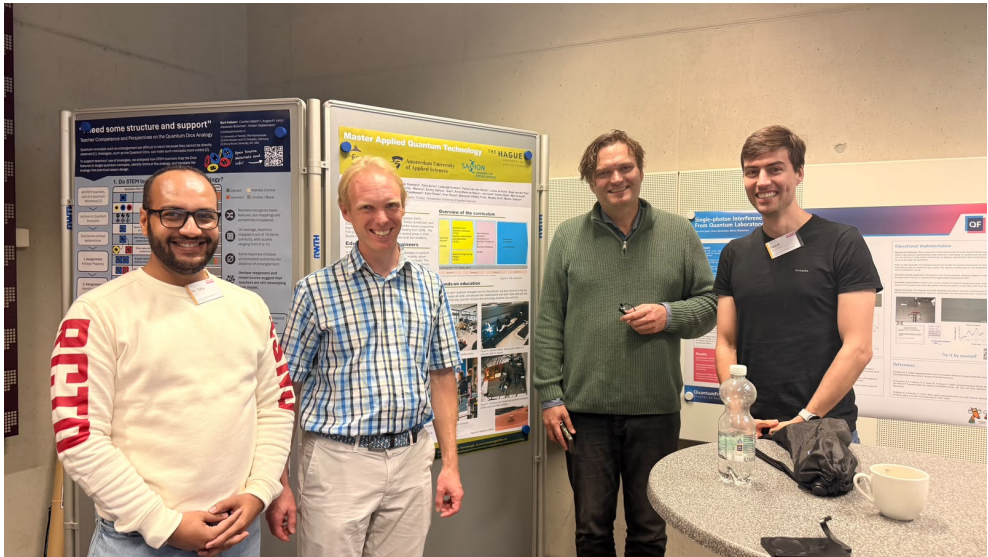
**Dr. Ir. Tjeerd
Bollmann**

Senior Lecturer and
Researcher, Saxion Uni-
versity of Applied Sciences

Dr. Taha Selim

Lecturer and Researcher,
Amsterdam University
of Applied Sciences

Part of the Applied QUantum Technology Master

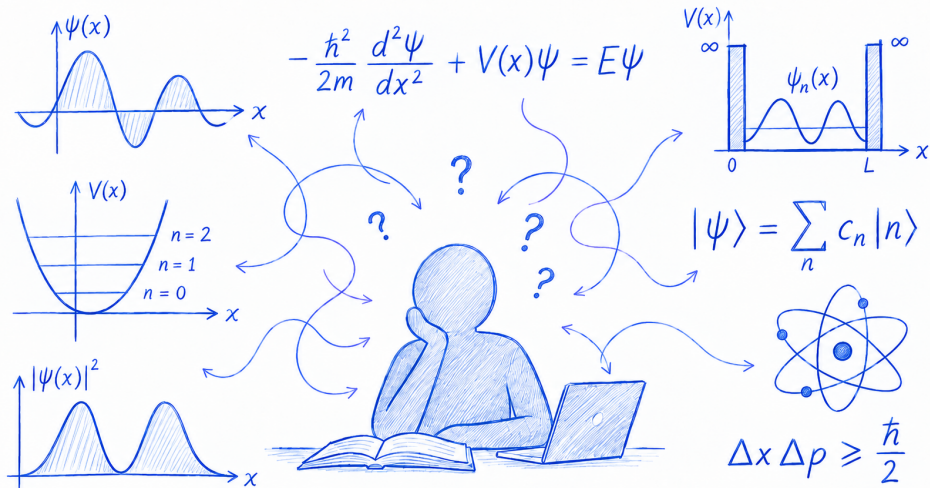


AI Act: Responsible Use of (Generative) AI in Education

Applied Quantum Technology Master — Draft Version 0.0

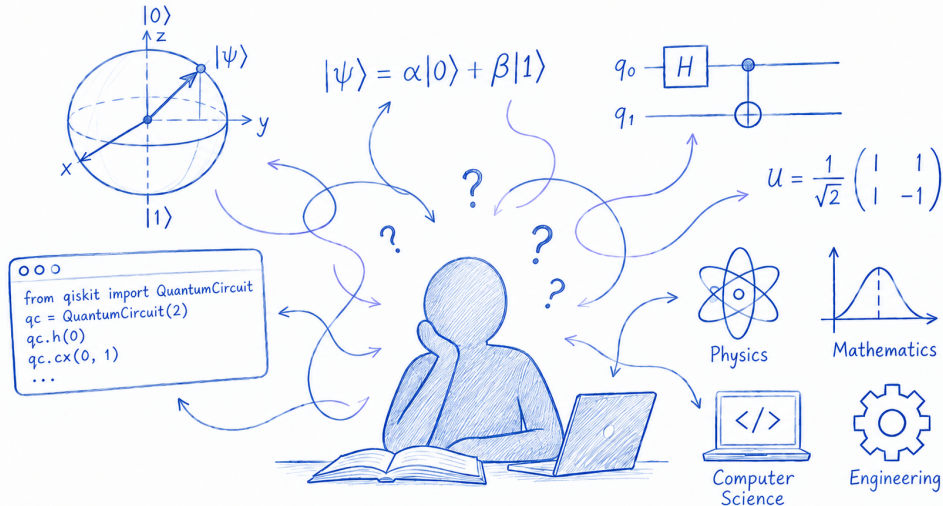
Why is quantum education difficult?

Quantum mechanics



Why is quantum education difficult?

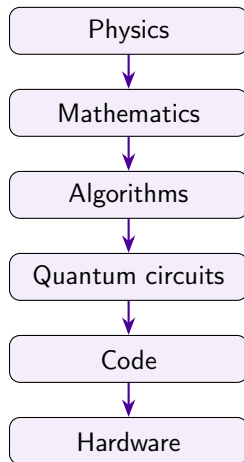
Quantum computing



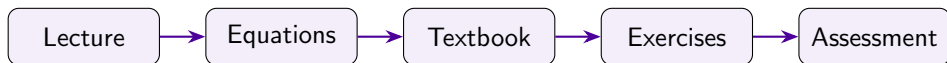
One learner, many representations

- Quantum concepts are abstract and often counterintuitive.
- Mathematical formalism appears very early.
- *Students arrive from physics, mathematics, computer science, and engineering to learn quantum computing.*
- Understanding requires moving fluently between several representations.

The challenge is not only learning each layer, but connecting them.



The traditional teaching workflow



Strengths

Structured progression, expert guidance, rigorous mathematics.

Typical constraints

One explanation for everyone, delayed feedback, static representations, limited individual support.

What can AI actually contribute?

Explain

adaptive explanations

Visualize

states, circuits, interference

Practice

exercises, hints, debugging

Reflect

critique, questioning, self-explanation

Students use AI as a tool for learning. Sometimes, They rely on AI for the whole work.

AI should reduce friction, not reduce thinking.

AI can give the illusion of understanding, It has to be supervised.

How do you use AI?
Let's collect some statistics.

Key considerations for AI in education

AI is already reshaping education

Use is growing

Teachers and students already use AI



Education is changing

AI is becoming part of learning



Our response matters

Adapt teaching with care and purpose

Principles for responsible classroom use

- 1 Keep learning first**
Support learning; do not replace student thinking.
- 2 Keep thinking active**
Protect reasoning, critical thinking and cognitive effort.
- 3 Make feedback useful**
Offer timely, specific feedback students can act on.
- 4 Build reflection**
Encourage self-explanation and verification of AI output.
- 5 Watch for over-reliance**
Notice when AI displaces independent work.

Help shape our educational AI framework.
What principles should guide it?

Productivity tool or substitute for learning?

Productive use of AI

- Clarifies difficult concepts
- Helps debug code
- Generates additional practice
- Visualizes abstract ideas
- Gives hints and rapid feedback

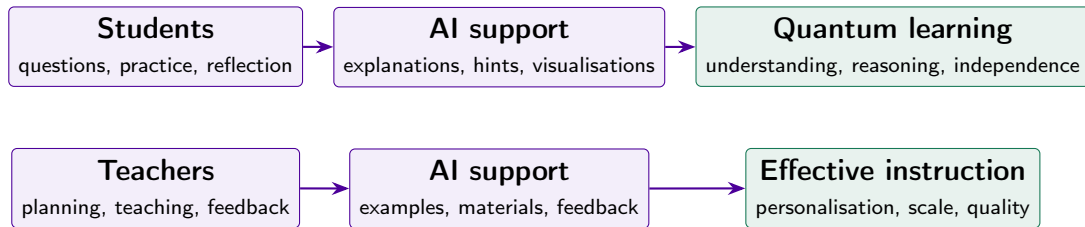
When AI hinders learning

- Gives answers without understanding
- Writes all the code
- Completes assignments
- Replaces mathematical practice
- Creates dependency

Does AI preserve the cognitive effort required for genuine understanding?

AI can be a productivity tool for teachers if used thoughtfully.

AI in the classroom



AI should strengthen learning and teaching, not replace them.

Scaffolding, not answer generation

How students should use it?

Poor use

“Solve this Schrödinger equation for me.”

Better use

“Do not give me the solution. Ask me one question at a time and give progressively stronger hints.”

Tutor mode

asks questions

Hint mode

progressive support

Verifier mode

critiques reasoning

Preserve productive struggle while removing unnecessary friction.

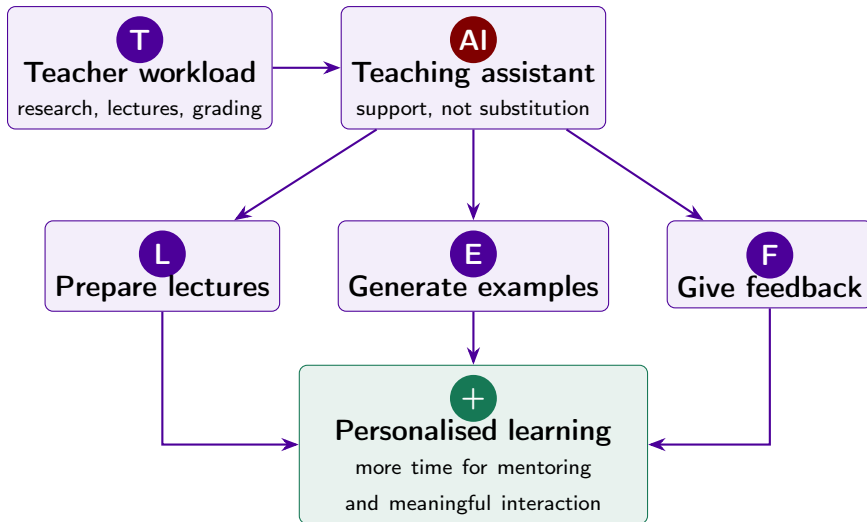
How teachers can use it?

- Teachers are often busy in research, preparing lectures, and grading assignments.
- AI can help in preparing lectures, generating examples, and providing feedback to students.
- AI can assist in creating personalized learning experiences for students.

How do you use it?

Scaffolding, not answer generation

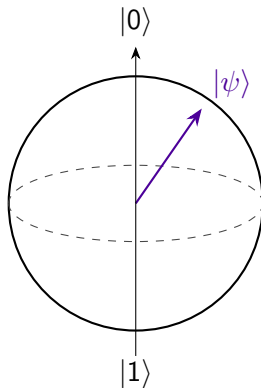
How teachers can use AI?



Example: superposition

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- 1 Ask for an intuitive explanation.
- 2 Visualize the state on a Bloch sphere.
- 3 Change α and β interactively.
- 4 Predict measurement probabilities.
- 5 Simulate measurements.
- 6 Compare prediction with observation.

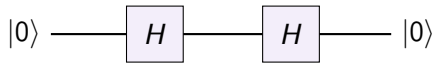


Prediction before simulation creates active learning.

Example: quantum interference

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H(H|0\rangle) = |0\rangle$$

Equation $\rightarrow$ Circuit $\rightarrow$ Visualization $\rightarrow$
Experiment $\rightarrow$ Explanation



Student task

Explain *why* two Hadamards return the original state before asking AI for an explanation.

AI-generated instructional media

AI can help instructors produce:

- short concept animations,
- narrated micro-lessons,
- Bloch-sphere demonstrations,
- circuit walkthroughs,
- mathematical derivations,
- pre-class revision material.

Learning cycle

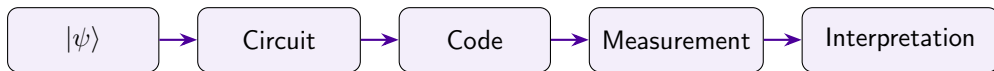
Before class: short concept video

During class: interactive simulation

After class: AI tutor + reflection

Reduce the instructor's production workload without reducing the student's intellectual workload.

From equation to circuit to code



```
qc.h(0)  
qc.cx(0, 1)
```

AI can translate between representations; students must understand the relationships between them.

AI-assisted programming: reason first, prompt second

Before using AI, students should:

- ① describe the intended quantum state,
- ② sketch the circuit,
- ③ predict the output.

Then AI can support:

- syntax and library usage,
- debugging,
- code explanation,
- troubleshooting,
- comparison of implementations.

Reason first. Prompt second.

Debugging as a learning exercise

Instead of asking AI to write the program, give students incorrect code:

```
qc.h(0)  
qc.cx(1, 0)
```

Ask:

- What state does this circuit produce?
- Why does it fail to create the intended Bell state?
- What change is required, and why?
- Can AI provide a *hint* rather than the correction?

The debugging process becomes the learning activity.

One-week university module

Day	Topic	AI-enhanced activity	Concept
1	Qubits & superposition	AI tutor + Bloch-sphere visualization	
2	Gates & measurement	Prediction + circuit simulation	
3	Entanglement	AI-guided Bell-state experiment	
4	Quantum algorithms	AI-assisted coding and debugging	
5	Mini-project	Design, implement, explain and defend	

→ **Predict** → **Build** → **Run** → **Explain** → **Reflect**

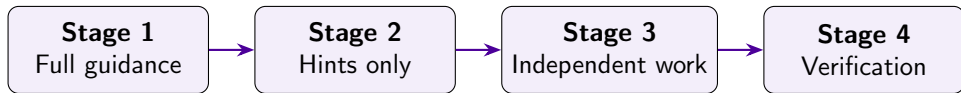
Example student workflow

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- 1 Explain the desired state.
- 2 Draw the circuit.
- 3 Predict measurement outcomes.
- 4 Implement and run it.
- 5 Ask AI to challenge the explanation.
- 6 Revise the reasoning.
- 7 Submit and defend the final explanation.

Predict → Build → Run → Explain → Defend

Progressive removal of AI scaffolding



Students should become less dependent on AI as their competence increases.

One classroom, different backgrounds

Physics student

Needs more support with programming and software tooling.

Computer-science student

Needs more support with physical interpretation and quantum formalism.

Engineering student

May need targeted support with linear algebra or abstraction.

Personalisation should adapt the path to understanding, not lower the level of understanding required.

What should we measure?

- Pre/post concept tests
- Error rates and code correctness
- Explanation quality
- Number of attempts
- Engagement and confidence
- Retention after several weeks
- Transfer to unfamiliar problems
- Mathematical reasoning
- Problem-solving time
- **Independent performance without AI**

A successful AI-supported course improves what students can do independently afterwards.

Where AI can hinder education

- Hallucinated physics
- Incorrect derivations or code
- Excessive trust in AI
- Reduced mathematical practice
- Cognitive offloading
- Loss of problem-solving persistence
- AI dependency
- Assessment integrity
- Privacy and institutional constraints

AI output is a hypothesis to verify, not an authority to accept.

Information delivery



Learning architecture + mentorship + verification

- design meaningful problems and learning trajectories,
- diagnose misconceptions,
- validate scientific correctness,
- connect theory to real applications,
- decide **when students should and should not use AI.**

Take-home message

- ① Quantum education requires students to connect physical intuition, mathematics, circuits and code.
- ② AI can provide scalable, personalised scaffolding for interdisciplinary cohorts.
- ③ AI should increase productivity **without replacing cognitive effort**.
- ④ Successful AI-supported learning should ultimately increase student independence.

The goal is not to create students who are better at using AI to solve quantum problems, but students who understand quantum systems more deeply because AI helped them learn.

Thank you

LinkedIn www.tahaselim.com

